Transportation of Data

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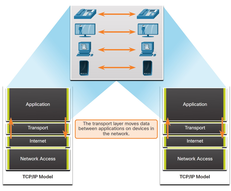
Role of the Transport Layer

Application layer programs generate data that must be exchanged between source and destination hosts. The transport layer is responsible for logical communications between applications running on different hosts. This may include services such as establishing a temporary session between two hosts and the reliable transmission of information for an application.

As shown in the figure, the transport layer is the link between the application layer and the lower layers that are responsible for network transmission.



The transport layer has no knowledge of the destination host type, the type of media over which the data must travel, the path taken by the data, the congestion on a link, or the size of the network.

The transport layer includes two protocols:

- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)

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Transport Layer Responsibilities

The transport layer has many responsibilities.

Select each tab for more information.

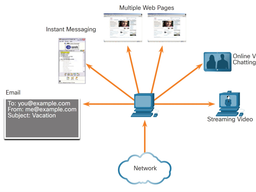
Tracking Individual Conversations | Segmenting Data and Reassembling Segments | Add Header Information | Identifying the Applications | Conversation Multiplexing

Tracking Individual Conversations

At the transport layer, each set of data flowing between a source application and a destination application is known as a conversation and is tracked separately. It is the responsibility of the transport layer to maintain and track these multiple conversations.

As illustrated in the figure, a host may have multiple applications that are communicating across the network simultaneously.

Most networks have a limitation on the amount of data that can be included in a single packet. Therefore, data must be divided into manageable pieces.



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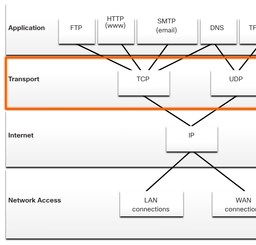
Transport Layer Protocols

IP is concerned only with the structure, addressing, and routing of packets. IP does not specify how the delivery or transportation of the packets takes place.

Transport layer protocols specify how to transfer messages between hosts, and are responsible for managing reliability requirements of a conversation. The transport layer includes the TCP and UDP protocols.

Different applications have different transport reliability requirements. Therefore, TCP/IP provides two transport layer protocols, as shown in the figure.



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Transmission Control Protocol (TCP)

IP is concerned only with the structure, addressing, and routing of packets, from original sender to final destination. IP is not responsible for guaranteeing delivery or determining whether a connection between the sender and receiver needs to be established.

TCP is considered a reliable, full-featured transport layer protocol, which ensures that all of the data arrives at the destination. TCP includes fields which ensure the delivery of the application data. These fields require additional processing by the sending and receiving hosts.

Note: TCP divides data into segments.

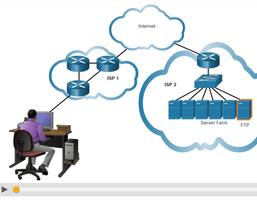
TCP transport is analogous to sending packages that are tracked from source to destination. If a shipping order is broken up into several packages, a customer can check online to see the order of the delivery.

TCP provides reliability and flow control using these basic operations:

- Number and track data segments transmitted to a specific host from a specific application
- Acknowledge received data
- Retransmit any unacknowledged data after a certain amount of time
- Sequence data that might arrive in wrong order
- Send data at an efficient rate that is acceptable by the receiver

In order to maintain the state of a conversation and track the information, TCP must first establish a connection between the sender and the receiver. This is why TCP is known as a connection-oriented protocol.

Click Play in the figure to see how TCP segments and acknowledgments are transmitted between sender and receiver.



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User Datagram Protocol (UDP)

UDP is a simpler transport layer protocol than TCP. It does not provide reliability and flow control, which means it requires fewer header fields. Because the sender and the receiver UDP processes do not have to manage reliability and flow control, this means UDP datagrams can be processed faster than TCP segments. UDP provides the basic functions for delivering datagrams between the appropriate applications, with very little overhead and data checking.

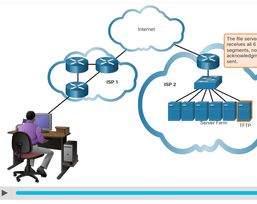
Note: UDP divides data into datagrams that are also referred to as segments.

UDP is a connectionless protocol. Because UDP does not provide reliability or flow control, it does not require an established connection. Because UDP does not track information sent or received between the client and server, UDP is also known as a stateless protocol.

UDP is also known as a best-effort delivery protocol because there is no acknowledgment that the data is received at the destination. With UDP, there are no transport layer processes that inform the sender of a successful delivery.

UDP is like placing a regular, nonregistered, letter in the mail. The sender of the letter is not aware of the availability of the receiver to receive the letter. Nor is the post office responsible for tracking the letter or informing the sender if the letter does not arrive at the final destination.

Click Play in the figure to see an animation of UDP datagrams being transmitted from sender to receiver.



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The Right Transport Layer Protocol for the Right Application

Some applications can tolerate some data loss during transmission over the network, but delays in transmission are unacceptable. For these applications, UDP is the better choice because it requires less network overhead. UDP is preferable for applications such as Voice over IP (VoIP). Acknowledgments and retransmission would slow down delivery and make the voice conversation unacceptable.

UDP is also used by request-and-reply applications where the data is minimal, and retransmission can be done quickly. For example, Domain Name System (DNS) uses UDP for this type of transaction. The client requests IPv4 and IPv6 addresses for a known domain name from a DNS server. If the client does not receive a response in a predetermined amount of time, it simply sends the request again.

For example, if one or two segments of a live video stream fail to arrive, it creates a momentary disruption in the stream. This may appear as distortion in the image or sound, but may not be noticeable to the user. If the destination device had to account for lost data, the stream could be delayed while waiting for retransmissions, therefore causing the image or sound to be greatly degraded. In this case, it is better to render the best media possible with the segments received, and forego reliability.

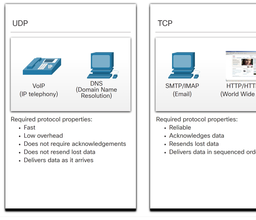
For other applications it is important that all the data arrives and that it can be processed in its proper sequence. For these types of applications, TCP is used as the transport protocol. For example, applications such as databases, web browsers, and email clients, require that all data that is sent arrives at the destination in its original condition. Any missing data could corrupt a communication, making it either incomplete or unreadable. For example, it is important when accessing banking information over the web to make sure all the information is sent and received correctly.

Application developers must choose which transport protocol type is appropriate based on the requirements of the applications. Video may be sent over TCP or UDP. Applications that stream stored audio and video typically use TCP. The application uses TCP to perform buffering, bandwidth probing, and congestion control, in order to better control the user experience.

Real-time video and voice usually use UDP, but may also use TCP, or both UDP and TCP. A video conferencing application may use UDP by default, but because many firewalls block UDP, the application can also be sent over TCP.

Applications that stream stored audio and video use TCP. For example, if your network suddenly cannot support the bandwidth needed to watch an on-demand movie, the application pauses the playback. During the pause, you might see a "buffering..." message while TCP works to re-establish the stream. When all the segments are in order and a minimum level of bandwidth is restored, your TCP session resumes, and the movie resumes playing.

The figure summarizes differences between UDP and TCP.



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Check Your Understanding - Transportation of Data

Check your understanding of the transport layer by choosing the correct answer to the following questions.

1. Which layer is responsible for establishing a temporary communication session between the source and destination host applications?
   - application layer
   - data link layer
   - network layer
   - physical layer
   - transport layer

2. Which three are transport layer responsibilities? (Choose three.)
   - [x] conversation multiplexing
   - identifying frames
   - identifying routing information
   - [x] segmenting data and reassembling segments
   - [x] tracking individual conversations

3. Which transport layer protocol statement is true?
   - TCP has fewer fields than UDP.
   - TCP is faster than UDP.
   - UDP provides reliability.
   - UDP is a best-effort delivery protocol.

4. Which transport layer protocol would be used for VoIP applications?
   - Session Information Protocol (SIP)
   - Transmission Control Protocol (TCP)
   - User Datagram Protocol (UDP)
   - VoIP Transfer Protocol

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